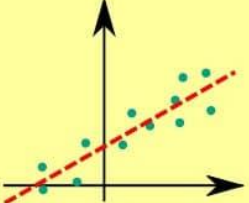
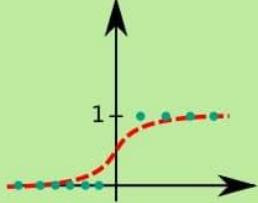
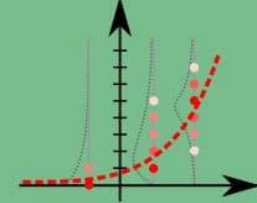


LINEAR REGRESSION	LOGISTIC REGRESSION	POISSON REGRESSION
① Econometric modelling ② Marketing Mix Model ③ Customer Lifetime Value	① Customer Choice Model ② Click-through Rate ③ Conversion Rate ④ Credit Scoring	① Number of orders in lifetime ② Number of visits per user
 Continuous $\Rightarrow$ Continuous $y = \alpha_0 + \sum_{i=1}^N \alpha_i x_i$ <code>lm(y ~ x1 + x2, data)</code> 1 unit increase in x increases y by α	 Continuous $\Rightarrow$ True/False $y = \frac{1}{1 + e^{-z}}$ $z = \alpha_0 + \sum_{i=1}^N \alpha_i x_i$ <code>glm(y ~ x1 + x2, data, family=binomial())</code> 1 unit increase in x increases log odds by α	 Continuous $\Rightarrow$ 0,1,2,... $y \sim \text{Poisson}(\lambda)$ $\ln \lambda = \alpha_0 + \sum_{i=1}^N \alpha_i x_i$ <code>glm(y ~ x1 + x2, data, family=poisson())</code> 1 unit increase in x multiplies y by e^α

MarketingDistillery.com is a group of practitioners in the area of e-commerce marketing.

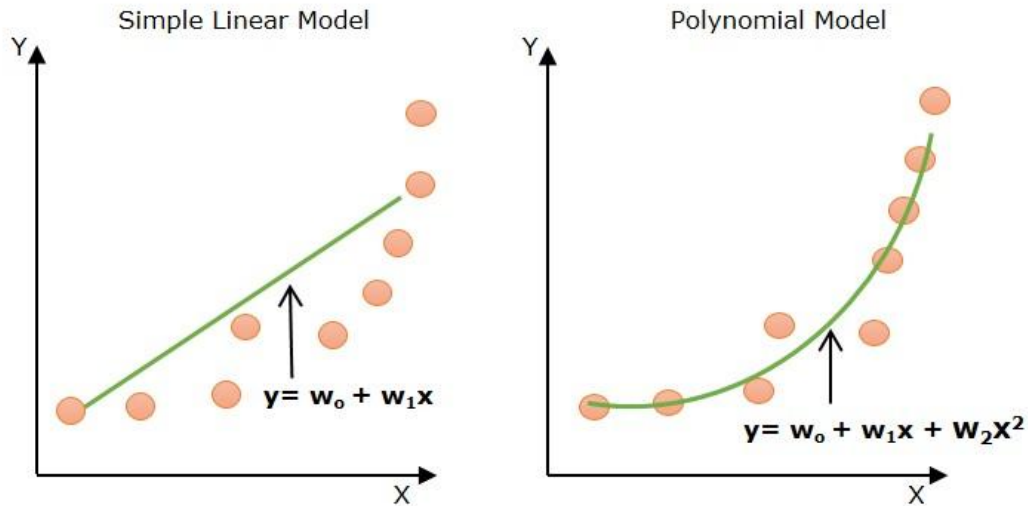


Advanced Regression Concepts

Bending the Rules and Taming the Algorithm

Instructor: Zaryab Ahmad Khan

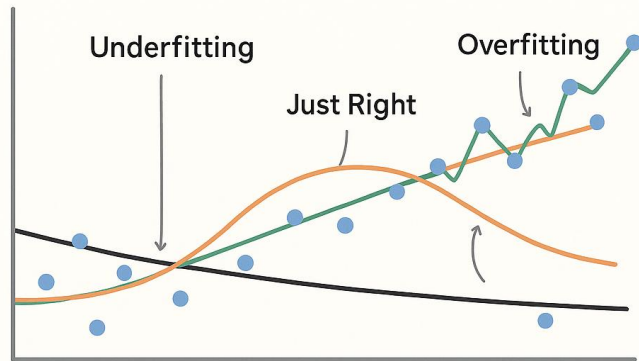
Polynomial Regression (Bending the Line)



- **The Problem:** Standard Linear Regression assumes data moves in a straight, predictable line. Real-world data (like viral spread, compound interest, or engine heat) often curves.
- **The Solution:** Polynomial Regression. We add "curves" to our math equation.
- **Concept:** Instead of just predicting based on X , we allow the model to learn from X^2 or X^3 , letting the line of best fit physically bend to follow the data points.

Overfitting vs. Underfitting (The Goldilocks Problem)

Overfitting vs Underfitting Explained Visually



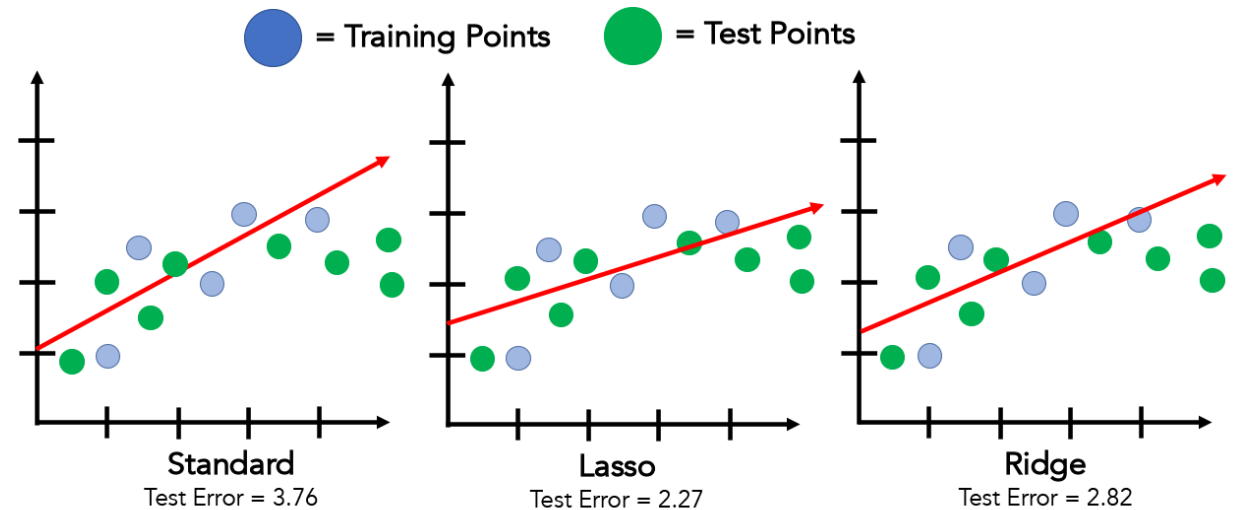
The goal is not to memorize the data, but to generalize the pattern.

Created by Mehul Ligade x.com/MehulLigade

- **Underfitting (Too Simple):** The model is a straight line trying to fit a curved dataset. It learns nothing useful and performs poorly on both training and testing data.
- **Overfitting (Too Complex):** The model connects every single dot perfectly. It has "memorized" the training data but will fail completely when given new, unseen data.
- **The Sweet Spot:** A model that captures the general trend without stressing over every tiny outlier.

Regularization (Ridge & Lasso)

- **The Concept:** Algorithms are greedy; they naturally want to overfit to get a "perfect" score. Regularization is a mathematical technique that gently "punishes" the model for getting too complex.
- **Ridge Regression (L2):** Shrinks the importance of less useful columns but keeps them in the equation.
- **Lasso Regression (L1):** The aggressive approach. It forces the importance of useless columns completely to zero, effectively deleting them and keeping the model simple.



**Our Data was actually "parabolic" but we couldn't tell from the small training sample.*